Alexander Kirilov



AI Engineer · Big Data Engineer · Team Lead

Java · Python · High-Load Streaming · Kafka · Kafka-connect CDC Expert · Kubernetes · AWS/GCP · Data Engineer for AI/ML Solutions

Email: kirilovu@gmail.com · **LinkedIn:** linkedin.com/in/kirilovu

SUMMARY

15+ years in production, four product companies, high-load streaming & ETL - reskilling into AI engineering over the past year on this Big Data background, Kafka Connect / CDC (Debezium) - expert, PostgreSQL - advanced, daily Claude Code.

Odnoklassniki (40M DAU, 350K req/sec, petabytes of data), Wargaming's ML/AI division (built the Java/Python/AWS SageMaker integration architecture powering their ML products - high-load infrastructure for AI systems before it was called that), Pfizer-level clinical platforms. Built and operated streaming data platforms end-to-end: Kafka / Kafka Connect, CDC (Debezium) - expert level, high-volume ETL into ClickHouse (analytics over 250M-record trial datasets, query time cut from minutes to seconds), advanced PostgreSQL (CDC, replication, performance tuning), Kubernetes-first delivery on AWS / GCP with monitoring and alerting (Prometheus, Grafana) built in. Experiment-driven rollouts at social-network scale: gradual releases with metric gates and runtime kill-switches.

Leadership & Business Impact: built products from scratch and formed the teams around them (led teams up to 10); worked directly with business and customers, defined scope and drove decisions. At Odnoklassniki extended the platform's game API and shifted notifications from mass to targeted - less spam across 140M users, higher notification response and player engagement, with BBox dynamically tuning per-app notification limits from per-user behaviour; multiple internal hackathon wins. At InData Labs worked closely with the customer and reshaped the product's payment model - grew payments and opened a new revenue channel. I own the full cycle - from customer conversations to shipping to production and keeping it stable.

AI Engineering - current focus: reskilling into AI engineering since January 2026 - completed a hands-on 7-month AI Engineer program, applying everything immediately in practice. A long-standing interest in psychology turned into Mirror (mirror.apps.aipush.dev) - an AI-therapy product I designed, built and run in production solo, integrating the modern AI stack: agentic orchestration, RAG, multi-agent pipelines, LLM evaluation and observability (details below). At my day job - daily AI-assisted engineering with Claude Code across the full SDLC. Next: delivering AI systems of this class for product and enterprise solutions, with 15+ years of production engineering discipline applied from day one.

CORE SKILLS

AI / GenAI: RAG pipeline design (incl. agentic RAG, GraphRAG) · Agentic systems & multi-agent orchestration (LangChain / LangGraph) · Automated prompt optimization (DSPy) · MCP: built and integrated custom MCP servers · Prompt engineering · LLM evaluation & guardrails (LLM-as-judge, A/B experiments) · Multimodal LLM apps (voice, image / handwriting OCR) · LLM observability & tracing · Vector search · Cost control

Self-hosted LLMs: Local LLM deployment and inference serving: hardware sizing, model customization with fine-tuning adapters (LoRA), custom models via distillation of larger ones

AI-assisted engineering: 1+ year of daily agentic development with Claude Code across the full SDLC - from Jira ticket to implementation, PR review, rollout plan and post-release metric monitoring · Built a Claude memory layer for a multi-service production system: architecture context maps, working rules, custom skills (incl. an architecture-check gate and a pre-review PR skill) shared in team GitHub repos

Big Data & Streaming: Kafka / Kafka Connect · CDC (Debezium) · High-volume ETL and streaming analytics pipelines · ClickHouse

Kafka Connect / CDC (Debezium) - expert: connector management, zero-downtime schema evolution, cross-version migration, WAL/sync-layer optimization at scale

High-load engineering: Production experience at a 40M DAU social platform (Odnoklassniki): experiment-driven delivery - gradual rollouts with metric gates and runtime kill-switches · custom in-house frameworks built for loads where standard stacks fail · platform services for anti-spam and game monetization APIs

Databases: PostgreSQL (advanced: CDC, replication, performance tuning) · ClickHouse (built a reporting system over 100M+ records on ClickHouse data fed via CDC from PostgreSQL) · MongoDB · Redis · Cassandra

Kubernetes & Docker: Kubernetes-first production architecture: designing, packaging and operating services for Kubernetes · Docker · Continuous integration / CD (GitHub Actions, Kustomize) · Production-ready operations: monitoring & alerting (Prometheus, Grafana)

Cloud: AWS · GCP

Languages & frameworks: Python (FastAPI, Pydantic, pytest) · Java, 15+ years (Spring Boot, Hibernate)

Leadership: Team lead (up to 10): formed and ran Scrum teams · Solution architecture: designed applications from scratch · Estimation & risk planning · Hiring and onboarding · Jira / Confluence

AI PRODUCT - MIRROR ("ZERKALO")

Founder / AI Engineer · 2026 - present · mirror.apps.aipush.dev

AI-powered psychological journal and therapist-matching platform (Telegram-first, FastAPI backend). Designed, built and operated solo, end-to-end - from architecture to a working product in production.

- **Agentic dialog orchestration:** LangGraph state machine with conversational moves, built-in crisis-detection safety exit, best-of-N reply selection.
- **Multi-agent pipelines:** therapist matching (client portrait → LLM-scored ranking → personalized explanation) and scheduled psychological-state reports.
- **RAG:** Qdrant vector search over parsed therapy literature, grounding persona replies in real case examples.
- **Multimodal input:** voice messages and handwritten notes via Gemini (speech recognition + OCR).
- **LLM evaluation:** LLM-as-judge with guard metrics (hallucinations, safety), simulated-client test harness (~300 scripted conversations), live A/B prompt experiments measuring retention and dialog depth; DSPy next in line.
- **Production ops:** Langfuse tracing, per-user LLM cost metering, Docker deployment with automated migrations.
- **Privacy-first design:** on-premise track - encryption, PII anonymization before external API calls, self-hosted LLM with LoRA adapters, self-hosted Whisper and OCR.

Stack: Python 3.12 · FastAPI · PostgreSQL 16 · SQLAlchemy · Qdrant · LangChain · LangGraph · cloud LLMs · Langfuse · Docker

Also: Generic RAG Agent - experimental web app: upload documents (PDF/TXT/MD), configure chunking and retrieval parameters, query them; my testbed for RAG parameter tuning.

EXPERIENCE

Senior Software Developer / Team Lead - REDCap Cloud

Dec 2021 - present · Astana, Kazakhstan (remote)

One of the world's leading platforms for clinical study data collection and management; clients include Pfizer and other global pharma companies. Led end-to-end design and delivery of a real-time clinical data analytics platform from the ground up.

- Built a streaming ETL pipeline - PostgreSQL CDC (Debezium) → Kafka Connect → ClickHouse - deployed on Kubernetes (AWS EKS); reduced analytics query time from minutes to seconds on trial datasets up to 250M records.
- Designed and built from scratch a connector-management service for the Kafka Connect / Debezium sync infrastructure: manages connectors across all environments and adds new tables to the sync without a full resync (previously 1-2 days of environment downtime); replaced ad hoc scripts and became part of the release process.
- Migrated the Kafka Connect / Debezium stack across major versions - reverse-engineered undocumented breaking changes, wrote the team's migration runbook.
- Optimized the PostgreSQL sync layer at scale: moved 75 tables off REPLICA IDENTITY FULL to cut WAL volume, added a materialized-view path for 50M-row tables (sync queries 60s → 20s), eliminated a 10GB-per-cycle temp-file spill.
- Architected and delivered a no-code reporting tool: web-based drag-and-drop interface (filters, groupings, drill-downs) letting clinical researchers build reports without SQL.
- Established the Kubernetes-first engineering standard (Docker, Kustomize, GitHub Actions CI/CD) and built a local Kubernetes dev environment running the full product stack on a single host - spinning up a complete environment now takes ~20 minutes, previously not possible at all.
- Led a multi-month cross-team investigation of user-authorization failures for a Pfizer-scale client across three repositories and two teams (CDC sync layer, OneLogin / Keycloak, analytics platform): root-caused three independent bugs and designed a single-source-of-truth auth architecture; separately caught a production memory leak three days before a major release.
- Started AI-assisted engineering on this project: Claude Code with a team memory layer (architecture context, custom skills in shared repos), MCP integrations, AI-driven incident investigation and analytics.
- Led the engineering team: formed the team, ran Scrum, owned architecture decisions, estimations and risk planning.

Stack: Java · Spring Boot · Kafka · Kafka Connect · Debezium · ClickHouse · Kubernetes · Kustomize · Docker · GitHub Actions · AWS

Technical Lead / Team Lead - InData Labs (Wargaming's ML/AI division)

Jun 2014 - Dec 2021 · Minsk

Big Data / ML / AI division of Wargaming. Started as a senior developer on Po.st, a high-load URL shortening and social sharing platform, then led backend teams (5-7 developers) across three AI products: NoginX (marketing automation), InfluencEye (influencer analytics), Skorebee (social scoring).

- Designed the ML/AI integration architecture connecting Python ML modules and AWS SageMaker into Java-based high-load services across all three products.
- Proposed and delivered a self-serve, token-based monetization system for InfluencEye - instant card payments instead of manual sales; the new revenue channel became a significant share of product revenue.
- Built data pipelines ingesting social platform APIs (Instagram, YouTube, TikTok, Facebook, Google Ads, Shopify) at scale, including sentiment analysis of influencer audience comments for demographic profiling.
- Built a company-wide dynamic configuration framework (DB-backed, live reload, change hooks) adopted across all projects for runtime feature toggles without redeploys.
- Earlier, on Po.st: data pipelines for link click statistics and user behaviour feeding the ad targeting engine of a top-4 US ad network - real-time aggregation at scale, spam detection, Facebook / Google Ads integrations.

Stack: Java · Spring Boot · Python · AWS (SageMaker, RDS, S3, SQS, SNS) · PostgreSQL · MySQL · MongoDB · Redis · Elasticsearch · RabbitMQ · Docker

Senior Software Developer - Odnoklassniki (ok.ru), Mail.Ru Group

May 2012 - Jul 2014 · Riga

One of the largest social networks in the world: 40M DAU, 350K req/sec, 650K media req/sec, 5,000+ servers, petabytes of data.

- Designed a smart notification targeting engine for social games: a platform API for game developers to tag and segment users (payment status, activity, custom markers) - enabled win-back campaigns and targeted offers; less spam across 140M users, higher notification response and player return rates.
- Designed and shipped "BBox", a viral rate-limiting engine regulating per-app quotas for feed posts and notifications based on per-user behavioral statistics - applying machine-learning concepts to platform-scale moderation before the AI era.
- Shipped features end-to-end: design → implementation → A/B experiment → performance analysis → rollout decision, on custom in-house infrastructure (proprietary network protocol, petabyte-scale distributed cache).
- Winner of internal hackathons: proposed and built messaging on top of then-brand-new HTML5 desktop notifications (message alerts while the browser is hidden), and animated avatars with recorded reactions to social-network actions.

Stack: Java · Spring · Cassandra · MSSQL · Memcache

Earlier career (2006 - 2012)

Senior Java Developer / Team Lead at **CrowdComputing Systems** (ML training-data pipelines on Amazon Mechanical Turk, ~1M labeling micro-tasks, clients incl. Microsoft Bing), **Exadel**, **SoftTeco**, **Qulix Systems**, **GP software.travel**. Founded **MAIKA.BY**, the first Belarusian custom t-shirt e-commerce (2005-2006).

EDUCATION & TRAINING

AI Engineer Course - 7-month hands-on program, 2026 (completed): agentic systems, RAG, LangChain / LangGraph, vector databases, MCP, multi-agent patterns, AgentOps & observability, agent evaluation, local LLM deployment and fine-tuning. Capstone: Mirror (above).

Belarusian State University of Informatics and Radioelectronics (BSUIR) - Diploma, Computer Science

LANGUAGES

English - B1 · Belarusian · Russian

INTERESTS

Djembe drumming (professional level) · Squash · Swimming · Psychology